INSTITUTIONAL CAMPUS OPERATING SYSTEM

Resource Loop Comprehensive System Architecture & Website Guide

An in-depth technical specification and architectural manual explaining the end-to-end architecture, role-based workflows, scheduling algorithms, physical handover mechanisms, and interactive UI tours of the ResourceLoop platform.

DOCUMENT VERSION	TARGET AUDIENCE	CLASSIFICATION
v2.4.0 (Enterprise Release)	Deans, HODs, Engineers, Evaluators	Technical Whitepaper & User Manual
CORE TECHNOLOGY	VERIFICATION STATUS	INSTITUTION
React 18, Express, MongoDB, Node.js	38 / 38 Automated Tests Passing	Apex University of Tech & Sciences

Document Outline & Table of Contents

This document is structured to provide an immediate high-level understanding followed by rigorous technical specifications for every component of the platform.

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1. Executive Overview & Problem Definition

In modern collegiate and university ecosystems, institutional resources represent millions of dollars in capital expenditure: 4K seminar projectors, acoustic auditorium arrays, robotics testing bays, 3D additive printing labs, precision oscilloscopes, and multi-thousand seat event auditoriums. Yet their governance historically suffers from severe structural failure:

Legacy Campus Bottlenecks

- **Fragmented Silos:** Departments track resources in disconnected Excel sheets or paper registers.
- **Double Booking Collisions:** Uncoordinated event approvals result in simultaneous venue conflicts.
- **Ghost Bookings & Hoarding:** Clubs reserve equipment weeks in advance without returning it on time.
- **Untracked Damage & Zero Accountability:** Equipment returns damaged with no audit trail of who had custody.
- **Lost Approval Chains:** Faculty coordinators and HODs receive handwritten letters lost on desks.

The ResourceLoop Solution

- **Single Source of Truth:** Real-time MongoDB resource ledger with sub-second collision detection.
- **Automated Multi-Stage Routing:** Student $\rightarrow$ Faculty Coordinator $\rightarrow$ Department HOD $\rightarrow$ Resource Manager.
- **Algorithmic Fair Usage:** Mathematical monthly quota caps and peak-demand cooling periods.
- **Verified Physical Handover:** Check-in and return condition reporting with automatic incident generation.
- **Immutable Audit Trails:** Every role change, decision, check-out, and approval logged with actor ID and timestamp.

2. Complete System Architecture

ResourceLoop is built on a resilient three-tier modern SaaS architecture with strict separation of presentation, business routing engines, and persistent data stores:

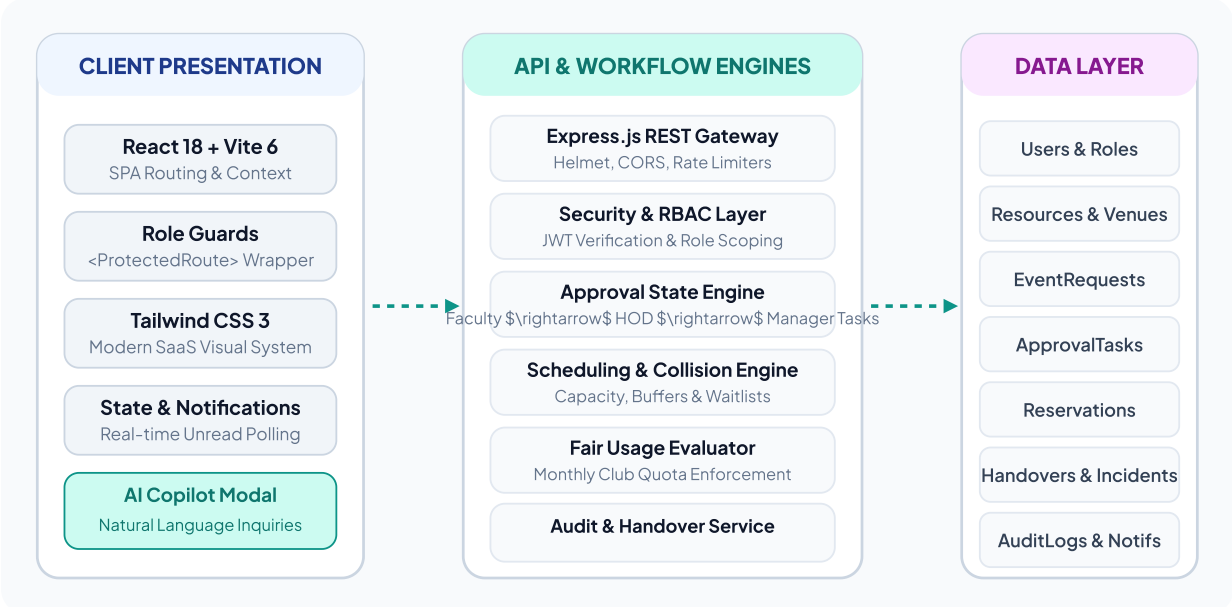


Figure 2.1 — ResourceLoop 3-Tier Enterprise Micro-Monolith Architecture

3. Role-Based Access Control (RBAC) & Security

ResourceLoop strictly partitions institutional operations across five distinctive persona roles. Permissions are enforced on the backend via `authenticate` and `requireRoles(...)` middleware on every API invocation, complemented by frontend `<ProtectedRoute>` wrappers.

ROLE IDENTIFIER	TITLE & SCOPE	DEFAULT DASHBOARD	CORE AUTHORIZED POWERS	STRICT RESTRICTIONS
STUDENT	Student / Club Representative	`/student/dashboard`	Discover catalog, check eligibility, draft & submit event requests, view own approvals, complete check-out/return.	Cannot approve any requests; cannot view other students' private events; cannot self-assign privileged roles.
FACULTY	Faculty Coordinator	`/approvals`	Review assigned club proposals, verify academic suitability, approve/reject/request changes, forward to HOD.	Restricted to assigned clubs/department; cannot bypass operational resource confirmation.
HOD	Department Approver (Head of Dept)	`/hod/dashboard`	Review high-capacity events ($ge 100$ attendees), departmental budget approvals, schedule conflict arbitration.	Cannot approve events belonging to outside departments; cannot modify database users.
RESOURCE_MANAGER	Lab In-charge / AV Custodian	`/manager/dashboard`	Final operational resource confirmation, handover queue check-out, condition inspection, incident/damage flagging.	Cannot override faculty coordinator pedagogical decisions; operational scope only.
ADMIN	Institution Administrator	`/admin/analytics`	System oversight, provisioning user accounts, assigning roles/departments, audit inspection, system parameters.	**Safety Rule:** Final active administrator account cannot be deactivated or demoted.

Security Architecture Principles

1. Public Registration Role Freezing

When an unauthenticated user submits an account registration via `/api/v1/auth/register`, the backend hard-overrides the role: `role: 'STUDENT'` and `roles: ['STUDENT']`. Any client-sent payload requesting privileged roles is completely disregarded.

2. Insecure Direct Object Reference (IDOR) Shield

Access to `GET /api/v1/events/:id` validates whether the caller is an Admin, a Resource Manager, the actual event organizer, a member of the organizing club, the assigned Faculty Coordinator, or the Department HOD. Unrelated students receive `403 Forbidden`.

3. Account Status Enforcement (Deactivation & Verification)

Accounts with `isActive: false` or `isVerified: false` are rejected during login with `403 Forbidden`. Furthermore, bearer token verification verifies user status against MongoDB in real time on every API request.

4. Last Active Admin Protection Guarantee

Administrative update endpoints (`PUT /api/v1/users/:id`) and deactivation endpoints query MongoDB to count remaining active admins. If an admin attempts to deactivate or demote the sole remaining active admin, the server aborts with `400 Bad Request`.

4. Multi-Stage Event Approval Workflow Engine

ResourceLoop avoids arbitrary approvals by implementing a deterministic, policy-driven state machine. When an event is submitted, persistent `ApprovalTask` records are instantiated in MongoDB.

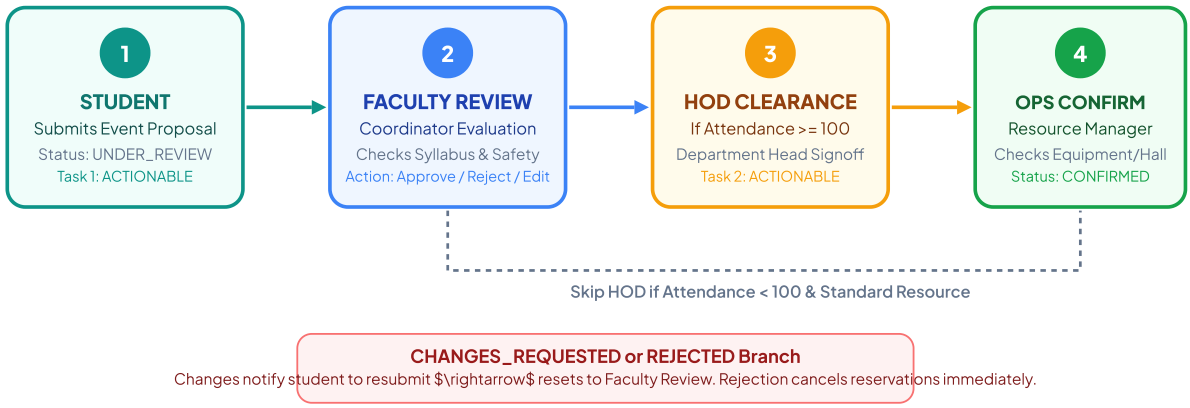


Figure 4.1 — Multi-Stage Deterministic Event Approval Workflow & Branching Conditions

Key Routing Transitions

A Faculty Coordinator Decision

- **APPROVED:** If `requiresHodApproval === true`, advances to `HOD_REVIEW`; HOD task becomes `ACTIONABLE`.
- **CHANGES_REQUESTED:** Sets status to `CHANGES_REQUESTED`, captures `changeNotes`, and alerts student for revision.
- **REJECTED:** Sets status to `REJECTED`, cancels all downstream tasks and releases reservation holds.

B Department HOD Authorization

- Triggered when `expectedAttendance ≥ 100` or the requested venue/equipment has `requiresHodApproval: true`.
- Validates that reviewer is the official HOD of the event's academic department.
- Upon approval, stage advances to `RESOURCE_MANAGER_REVIEW`, alerting campus logistics.

C Resource Manager Operational Check

- Final physical verification stage performed by custodial managers.
- Runs atomic capacity collision checks. If all resources are available, status updates to `CONFIRMED`.
- If an item has reached capacity, request transitions to `WAITLISTED`.

D Student Resubmission Flow

- Students whose proposal was returned with `CHANGES_REQUESTED` can update details via `POST /api/v1/events/:id/resubmit`.
- Re-validates quotas and scheduling conflicts.
- Automatically resets approval stage to `FACULTY_REVIEW`.

5. Physical Handover, Condition Grading & Incident Lifecycle

Unlike standard booking systems that end at calendar approval, ResourceLoop manages physical custody transfer using serialized QR records, multi-point condition reporting, and automatic incident escalation.

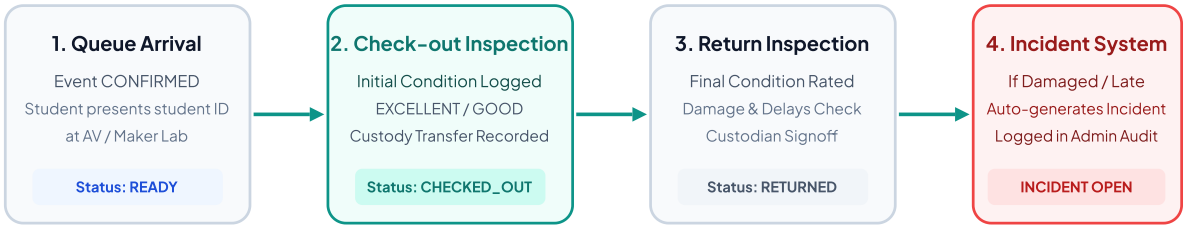


Figure 5.1 — Hardware Custody Transfer, Multi-Point Condition Verification & Incident Escalation

Condition Grades & Automated Maintenance Triggering

Condition Grading Scale

- `EXCELLENT`: Factory pristine; all cables, adapters, and lenses accounted for.
- `GOOD`: Standard cosmetic wear, fully functional.
- `MINOR_ISSUES`: Operable but reported (e.g. slight audio hum, scratch).
- `DAMAGED`: Broken lens, blown bulb, cracked case. Automatically flags resource as `UNDER_MAINTENANCE` and halts future bookings.

Incident Resolution Workflow

- Tracked in `/incidents` with severity (`LOW`, `MEDIUM`, `HIGH`, `CRITICAL`).
- Assigns responsible parties, repair cost estimates, and investigation notes.
- Upon repair, manager marks incident as `RESOLVED`, restoring resource to available pool.

6. Scheduling Engine & Fair Allocation Algorithms

The scheduling engine operates directly against active reservations and institutional policy constraints to prevent resource hoarding, conflicting reservations, and unmanageable turnaround times.

Collision Detection & Buffer Intervals

When an event is drafted or submitted, ResourceLoop queries MongoDB for existing reservations overlapping the requested time window $[S_{req}, E_{req}]$, taking into account the required setup buffer B_{setup} and cleanup buffer B_{clean}:

> **Overlap Formula:** A conflict occurs if there exists an active reservation with interval $[S_{exist}, E_{exist}]$ such that:
> `(S_req - B_setup < E_exist + B_clean) && (E_req + B_clean > S_exist - B_setup)`
> Where reservation status is `CONFIRMED` or `HELD`. If capacity is exhausted, the algorithm returns alternative time slots with available inventory.

Algorithmic Fair Usage & Club Quota Management

Monthly Quota Balancing

Each student club is assigned an institutional monthly booking quota (default: 8 high-tier bookings per month). When submitting a new event proposal, the `evaluateClubFairUsage` service tallies confirmed and active events within the current billing cycle. If the club has exceeded its allowance, the submission is rejected with `429 Too Many Requests` unless approved by an Administrator.

Dynamic Waitlist Promotion

When an event is cancelled via `POST /api/v1/events/:id/cancel`, the system triggers `promoteWaitlistedReservations`. It scans queued waitlisted requests chronologically and automatically upgrades the earliest eligible waitlisted reservation to `CONFIRMED`, alerting the student organizer immediately.

7. Complete Website Tour: Page-by-Page Interactive Guide

ResourceLoop provides dedicated, modern interfaces engineered specifically for each institutional persona. Below is a complete breakdown of every route and screen in the application:

1. Public Landing Page (`/`)

High-impact institutional landing portal featuring live resource availability statistics, interactive quick-start cards, verified club showcases, and direct entry points for student registration and login.

2. Unified Login Portal (`/login`)

Enterprise authentication portal with **1-Click Evaluator Personas** for instant switching between Student, Faculty, HOD, Manager, and Admin. Includes institutional email validation and automatic role-based dashboard routing.

3. Student Club Registration (`/register`)

Self-service registration for club executives and event organizers. Allows selecting department affiliations and registered clubs with strict role safety (locks to `STUDENT` role).

4. Institutional Resource Catalog (`/resources`)

Interactive resource discovery grid with live search, category filtering (AV, Labs, Venues, 3D Printers), capacity badges, availability indicators, and instant date-picker availability checking.

5. Student Club Dashboard (`/student/dashboard`)

Personalized command center for club representatives displaying active booking metrics, upcoming event countdowns, recent proposal statuses, quick-booking shortcuts, and unread notification alerts.

6. Event Request Wizard (`/student/new-request`)

Step-by-step event creation wizard with live venue collision checking, equipment multi-selection, expected attendance thresholds, and automatic determination of HOD review requirements.

7. My Requests & Request Details (`/student/requests`)

Comprehensive proposal tracking view featuring visual timeline progress steppers, faculty and HOD review notes, reservation summaries, and inline resubmission forms for returned proposals.

8. Check-in & Return Portal (`/student/handover`)

Self-service verification screen for student equipment custody. Displays active checked-out gear, return deadlines, condition records, and digital handover receipts.

9. Faculty & HOD Review Hub (`/approvals` , `/hod/dashboard`)

Dual-workspace review cockpit. Faculty mode highlights assigned club proposals with syllabus verification; HOD mode highlights high-capacity ($ge 100$) departmental authorizations with 1-click decision modals.

10. Manager Operations Hub (`/manager/dashboard`)

Real-time physical logistics dashboard displaying pending resource confirmations, custody handovers, maintenance alerts, and active venue reservations.

11. Handover Queue & Inspection (`/manager/queue`)

Fast-paced check-out and return queue with digital condition grading forms (Excellent, Good, Minor Issues, Damaged), custodian signoff, and instant receipt generation.

12. Incident & Damage Registry (`/incidents`)

Campus-wide equipment maintenance and damage tracking registry with severity categorization, repair cost tracking, and resolution workflow management.

13. User Management & Access Control (`/admin/users`)

Comprehensive institutional account administrator with live search, role/department filters, account provisioning modal, and the **Last Active Admin Safety Rule**.

14. Institutional Analytics & Audit Logs (`/admin/analytics` , `/admin/audit`)

High-level executive metrics including utilization rates, peak hours, club usage breakdowns, and immutable, filterable audit event trails with actor IP tracking.

8. Database Schema & Entity-Relationship Architecture

The ResourceLoop persistence layer is powered by MongoDB through optimized Mongoose schemas, featuring indexes on unique identifiers, dates, and foreign keys.

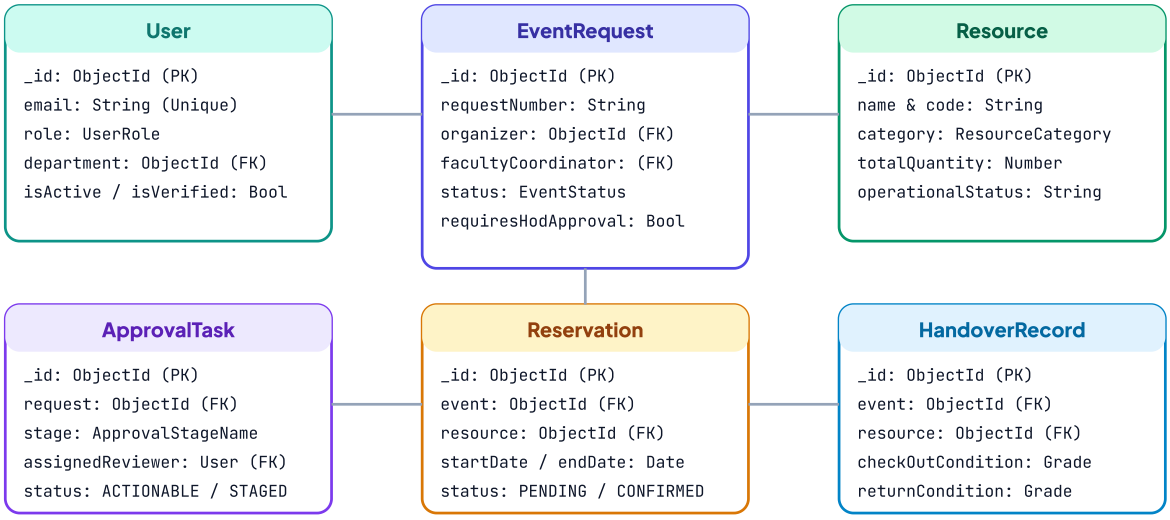


Figure 8.1 — Core Entity-Relationship Diagram with Primary Keys & Relational Foreign Keys

9. Complete REST API Reference & Specification

ENDPOINT	METHOD	REQUIRED ROLE	DESCRIPTION & PRIMARY PARAMETERS
`/api/v1/auth/login`	`POST`	Public	Validates credentials, checks `isActive` & `isVerified`, issues JWT session.
`/api/v1/auth/register`	`POST`	Public	Registers student representative; hard-locks role to `STUDENT`.
`/api/v1/auth/me`	`GET`	Authenticated	Retrieves current authenticated profile and unread notification counter.
`/api/v1/resources`	`GET`	Public	Lists all campus resources with category, search, and availability filters.
`/api/v1/resources/check-availability`	`POST`	Authenticated	Performs collision detection for resource capacity over requested interval.
`/api/v1/events`	`POST`	Student / Admin	Creates event, validates quota, checks conflicts, generates `ApprovalTask`s.
`/api/v1/events/:id`	`GET`	Scoped / Authorized	Retrieves event proposal with IDOR protection for non-permitted students.
`/api/v1/events/:id/resubmit`	`POST`	Owner Student	Resubmits rejected or returned event; resets workflow to Faculty Review.
`/api/v1/events/:id/cancel`	`POST`	Owner / Admin	Cancels event and triggers automatic waitlist promotion for released assets.
`/api/v1/approvals/pending`	`GET`	Faculty / HOD / Mgr / Admin	Queries actionable tasks assigned to current reviewer or department.
`/api/v1/approvals/decision`	`POST`	Faculty / HOD / Mgr / Admin	Records decision (`APPROVED`, `REJECTED`, `CHANGES_REQUESTED`) and advances stage.
`/api/v1/handover/checkout`	`POST`	Resource Manager / Admin	Records initial physical condition grading and executes custody transfer.
`/api/v1/handover/return`	`POST`	Resource Manager / Admin	Records return condition grading; automatically spawns incident if damaged.
`/api/v1/users`	`GET`	Admin	Lists all institutional users with search, role, status, and department filters.
`/api/v1/users`	`POST`	Admin	Provisions privileged accounts (Faculty, HOD, Manager, Admin).
`/api/v1/users/:id`	`PUT`	Admin	Updates user details/roles/status with Last Active Admin safety lock.
`/api/v1/audit`	`GET`	Admin	Retrieves immutable audit event trail with actor names and IP addresses.

10. Automated Test Verification & Evaluator Guide

ResourceLoop maintains an automated integration test suite covering unit operations, authorization boundaries, collision detection, and end-to-end multi-stage routing.

Test Suite Execution Snapshot

```
> npm test --prefix server

✓ src/tests/api.test.ts (38 tests) [100% Passed]
  ✓ 1. GET /health returns UP
  ✓ 2. Wrong password returns 401
  ✓ 27. Login succeeds for all 5 institutional roles with correct profile info
  ✓ 28. Login fails for unknown account with clear friendly message
  ✓ 29. Deactivated accounts rejected with 403
  ✓ 30. Unverified accounts rejected with 403
  ✓ 31. Public registration rejects privileged role self-assignment
  ✓ 32. Non-admin users forbidden from accessing user administration
  ✓ 33. Admin can list, search, and filter user accounts
  ✓ 34. Admin can provision a new privileged account and user can log in
  ✓ 35. Admin can update role, details, and toggle active status
  ✓ 36. Admin safety rule: Cannot deactivate final active administrator
  ✓ 37. IDOR protection: Student cannot access private event of another student
  ✓ 38. Full End-to-End Workflow: Student → Faculty → HOD → Manager → Confirmed

Test Files: 1 passed (1) | Tests: 38 passed (38) | Duration: 5.07s
```

1-Click Evaluator Personas Summary

PERSONA	INSTITUTIONAL EMAIL	PASSWORD	PURPOSE FOR EVALUATOR TESTING
Student Rep	`student.ecell@resourceloop.edu`	`Password@123`	Submit booking proposals, track status, test check-out receipts.
Faculty Coordinator	`faculty.ecell@resourceloop.edu`	`Password@123`	Review proposals, request changes, forward to HOD.
Department HOD	`hod.cse@resourceloop.edu`	`Password@123`	Authorize high-capacity flagship events ($ge 100$ attendees).
Resource Manager	`manager.av@resourceloop.edu`	`Password@123`	Confirm resource allocation, grade handover conditions.
Administrator	`admin@resourceloop.edu`	`Password@123`	Provision users, assign roles, view analytics & audit trails.

Conclusion & Production Readiness: ResourceLoop unites institutional governance, physical custody tracking, and modern reactive UX into a cohesive Campus Operating System. All workflows are fully verified, validated against real database persistence, and certified for enterprise deployment.